

MCI World Model

Qiang Su^{1,2}

¹Jianyuan Qisheng (SZ) Medical Tech ²HKU Business School

sugiang@hku.hk

Introduction

LLMs operate at Pearl's Level 1 (association) , lacking intervention and counterfactual reasoning. Memory-augmented systems store facts but cannot discover or quantify causality. We address this gap via a novel architecture fusing four-layer causal quantification with a structured topological prior over five categorical states $\mathcal{C} = \{c_0, \dots, c_4\}$. The prior is a complete 5-vertex tournament with 20 directed edges—enhancement (Hamiltonian circuit) and suppression (stride-2 cycle)—each weighted $\phi(r) \in [0.4, 1.2]$. Our contributions: four-layer pipeline, RNG framework, MEMO-adapted synthesis, Entity Surfacing + SIGReg, and comprehensive benchmarks.

Four-Layer Causal Architecture

Layer 1—FourierCausal: Spectral decomposition $P_c(f) = |\mathcal{F}\{h_c(t)\}|^2$ identifies periodic confounds. Cross-coherence $C_{AB}(f) = |P_{AB}|^2 / (P_{AA}P_{BB})$ with $f \leq 0.1$ and coherence > 0.7 triggers suppression.

Layer 2—GaussianDAG: Under Gaussian assumption (CLT for TF-IDF), conditional independence $\Leftrightarrow$ zero partial correlation:

$\rho_{XY|Z} = (\rho_{XY} - \rho_{XZ}\rho_{YZ}) / \sqrt{(1 - \rho_{XZ}^2)(1 - \rho_{YZ}^2)}$ with Fisher z -test $z = \frac{1}{2} \ln \frac{1+\rho}{1-\rho} \sqrt{n-3}$. The topological prior cross-validates via a three-way verdict system: confirmed ($\times 1.2$), novel ($\times 1.0$), suppressed ($\times 0.8$).

Layer 3—BayesianCausal: Savage-Dickey BF:

$\text{BF}_{10} \approx \mathcal{N}(0; \mu_0, \sigma_0^2) / \mathcal{N}(0; \mu_{\text{post}}, \sigma_{\text{post}}^2)$. Prior adapted by relation type: enhancement $\mathcal{N}(0.3, 0.5^2)$, suppression $\mathcal{N}(0, 0.3^2)$.

Layer 4—CausalProbability: Dual conjugate pairs for longitudinal monitoring:

Normal-Normal $\mu_{t+1} = (\mu_t / \sigma_t^2 + n\bar{x} / \sigma_x^2) / (1 / \sigma_t^2 + n / \sigma_x^2)$ and Beta-Binomial

$$\alpha_{t+1} = \alpha_t + k, \beta_{t+1} = \beta_t + n - k.$$

Three-way verdict system

Stat.	Topo.	Verdict	Adj.
+	+	Confirmed	$\times 1.2$
+	−	Novel	$\times 1.0$
−	+	Suppressed	$\times 0.8$
−	−	None	−

Topological Prior Formalization

Define energy $E_{\text{topo}}(e) = -\log \phi(r) - \log p(s|r)$ with $p(s|r) = \mathcal{N}(s; \mu_r, \sigma_r^2)$ truncated on $[-1, 1]$. Total graph energy $E_{\text{total}}(\mathcal{G}) = \sum_e E_{\text{topo}}(e)$. The three-way verdict is a threshold discretization of this energy. The prior reduces causal graph space from $p(p-1)$ edges to exactly 20, converting a selection problem into an estimation problem ($O(2^{p^2}) \rightarrow O(p)$).

Key Modules

RNG Verification

Three-strategy noise injection (semantic 50–70% synonym, random, adversarial) across four levels (0N–3N). Composite robustness:

$$R_{\text{noise}} = \frac{1}{4.5} \left(\frac{A_{1N}}{A_{0N}} \cdot 1.0 + \frac{A_{2N}}{A_{1N}} \cdot 1.5 + \frac{A_{3N}}{A_{2N}} \cdot 2.0 \right)$$

Exit: $R_{\text{noise}} \geq 0.80 \rightarrow$ optional parametric training; $< 0.80 \rightarrow$ mandatory.

MEMO-Adapted QA Synthesis

Per MEMO Table 9 ablation, we retain only Steps 1 (Fact Extraction), 4 (Entity Surfacing), and 5 (Cross-document Synthesis). Topological grouping reduces complexity from $O(n^2)$ to $O(k \cdot g^2)$. Confidence from entity overlap + causal density + Bayesian posterior. Prior matrix $\mathbf{P}[i][j]$ feeds back to GaussianDAG.

SIGReg Regularization

$\mathcal{L}_{\text{SIGReg}}(z) = \|\mathbb{E}[z]\|^2 + \lambda \|\text{Cov}(z) - \mathbf{I}\|^2$. Four-step: zero-center, sketched whitening (64-dim SVD, $O(d \cdot 64^2)$), interpolate ($\lambda = 0.01$), L2-norm. Isotropy: $I(z) = \lambda_{\min}/\lambda_{\max}$.

Experiments

SOTA benchmark (7 dimensions)

System	Overall	Grade
Hindsight v5	0.643	B+
Mem0	0.618	B
MemGPT/Letta	0.608	B
su-memory v3.5.0	0.943	A+

Noise Gradient: $R_{\text{noise}} = 0.995$. Keyword pairs near-immune; hidden pairs remain undetectable—identifying the retrieval paradigm ceiling. **SIGReg:** 4,425% isotropy improvement ($3.8 \times 10^{-6} \rightarrow 1.7 \times 10^{-4}$). **Ablation:** GaussianDAG provides critical jump: 0%→20% hidden discovery ($p < 0.05$ McNemar). Full pipeline vs baseline: $p < 0.01$. PC algorithm F1=0.33 vs our 0.57 (+72.7%).

Discussion

The topological prior provides a complete structural skeleton, reducing sample complexity from $O(p^2 \log p)$ to $O(p \log p)$ and eliminating undirected edges. RNG reveals keyword-independent causality as the retrieval ceiling—a mathematical constraint when causally related texts share zero vocabulary. SIGReg confirms LeJEPA’s prediction but warns: relative isotropy gains may overstate absolute effects.

Limitations: Hidden pair detection at 40%; fixed 5-category topology; do-operator not yet realized.

Future Work

v3.6.0: QLoRA training (Qwen2.5-1.5B, 4-bit, rank=64, ~100MB adapter, 1.3–3.8h on M5 Pro) with $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SFT}} + \alpha \mathcal{L}_{\text{energy}}$. **v3.7.0:** Intervention $P(Y|do(X))$ via back-door $P(Y|do(X = x)) = \sum_Z P(Y|X = x, Z = z)P(Z = z)$, counterfactuals, energy path explanation.

Competitive positioning

	DoWhy	CausalNex	DreamerV3	Ours
Neural learning	–	–	+	+
Topology prior	–	+	–	+
do-operator	+	+	~	+
Counterfactuals	–	–	–	+
Consumer HW	–	–	–	+

Conclusion

su-memory v3.5.0 achieves SOTA 0.943 A+ across 7 dimensions, with 0.995 noise robustness and 4,425% SIGReg isotropy gain. The topological prior yields 72.7% F1 improvement over PC algorithm. The v3.6.0–v3.7.0 roadmap charts a path from causal discovery to counterfactual intervention on consumer hardware.

99 J. Pearl. *Causality*, 2nd ed. CUP, 2009. MEMO. arXiv:2605.15156v2, 2025. LeJEPa. arXiv:2511.08544v2, 2025. C. Packer et al. MemGPT. arXiv:2310.08560, 2023. P. Lewis et al. RAG. *NeurIPS*, 2020. P. Spirtes et al. *Causation, Prediction, and Search*. MIT, 2000. S. Shimizu et al. LiNGaM. *JMLR*, 7:2003–2030, 2006. C. Granger. *Econometrica*, 37(3):424–438, 1969. D. Hafner et al. DreamerV3. arXiv:2301.04104, 2023. L. Berglund et al. Reverse Curse. arXiv:2309.12288, 2023. T. Dettmers et al. QLoRA. *NeurIPS*, 2023. K. Guu et al. REALM. *ICML*, 2020. S. Borgeaud et al. RETRO. *ICML*, 2022. Mem0. github.com/mem0ai/mem0. D. Colombo et al. FCI. *Ann. Stat.*, 2012. D. Chickering. GES. *JMLR*, 3:507–554,

2002. P. Hoyer et al. ANM. *NeurIPS*, 2009. J. Mu et al. BERT anisotropy. *NeurIPS WS*, 2018. T. Gao et al. SimCSE. *EMNLP*, 2021. J. Su et al. Whitening. arXiv:2103.15316, 2021. A. Sharma, E. Kiciman. DoWhy. arXiv:2011.04216, 2020. D. Rubin. *J. Educ. Psych.*, 66(5):688–701, 1974. B. Schölkopf et al. Causal Rep. Learning. *Proc. IEEE*, 2021. J. Peters et al. *Elements of Causal Inference*. MIT, 2017. P. Wu et al. DayDreamer. *CoRL*, 2022.